Passwords Obtained:
Bob: W1nt3rC0c0a
Charlie: AutumnL3av3s
Diana: S3cur3L0ck
Edward: H3ll0Gr33t
Frank: R0ck3tLaunch
Grace: Fr3shBr1ght
Irene: FunT1m3
Karen: S4f3Checks
Leo: 3Rul35
Mason: $Succ3ss

Missed:

- Alice
- Henry
- Jack
- Monica
- Admin

Workflow
:
c
sa@ se3 si1 so0 ss5 c sa@ c se3 c si1 c so0 c ss5 c sa4 c s3e c s0o c s1i c s@a c s4a
These are the rules that were applied in my cust.rule file

Base Bob dictionary:

Winter
W1nter
wint3r
Night
n1ght
Latenight
L4tenight
l@tenight
laten1ght
Gaming
G4ming

g@ming
gam1ng
Cocoa
C0coa
Coc0a
Coco4
coco@

// hashcat -a 0 bob.txt -r cust.rule --stdout > bob_stage1.txt:

- Applies all rules to the base file

hashcat -a 1 --stdout bob_stage1.txt bob_stage1.txt > bob2.txt

- Appends all combination of each string and leet variation

hashcat -m 1800 -a 0 --username bob.hash bob2.txt

This attacks the password with the modified dictionary

hashcat --show bob.hash

shows final password

```
┌──(kali㉿kali)-[~/passwordHomework/password_files_2]
└─$ hashcat -a 0 bob.txt -r cust.rule --stdout > bob_stage1.txt

┌──(kali㉿kali)-[~/passwordHomework/password_files_2]
└─$ hashcat -a 1 --stdout bob_stage1.txt bob_stage1.txt > bob2.txt

┌──(kali㉿kali)-[~/passwordHomework/password_files_2]
└─$ hashcat -m 1800 -a 0 --username bob.hash bob2.txt
hashcat (v6.2.6) starting

OpenCL API (OpenCL 3.0 PoCL 5.0+debian  Linux, None+Asserts, RELOC, SPIR, LLVM 17.0.6, SLEEF, DISTRO, POCL_DEBUG) - Platform #1 [The pocl project]
====================================================================================================================================
* Device #1: cpu-penryn-QEMU Virtual CPU version 2.5+, 14996/30057 MB (4096 MB allocatable), 6MCU

Minimum password length supported by kernel: 0
Maximum password length supported by kernel: 256

INFO: All hashes found as potfile and/or empty entries! Use --show to display them.

Started: Mon Jan 26 23:51:10 2026
Stopped: Mon Jan 26 23:51:10 2026

┌──(kali㉿kali)-[~/passwordHomework/password_files_2]
└─$ hashcat --show bob.hash
Hash-mode was not specified with -m. Attempting to auto-detect hash mode.
The following mode was auto-detected as the only one matching your input hash:

1800 | sha512crypt $6$, SHA512 (Unix) | Operating System

NOTE: Auto-detect is best effort. The correct hash-mode is NOT guaranteed!
Do NOT report auto-detect issues unless you are certain of the hash type.

$6$Dew18.zhdfJS3EmM$JBbr5D4gsekbaVbvDKphMBVEG3UQyvBL498/kVNOG/Rp0xu2YJLH9lnP.bfoHDGPy.6MUz85hbJH8ypoDVbGx/:W1nt3rC0c0a

┌──(kali㉿kali)-[~/passwordHomework/password_files_2]
└─$
```